

MacSet-WX1 Wi-Fi IoT Relay Controller Datasheet

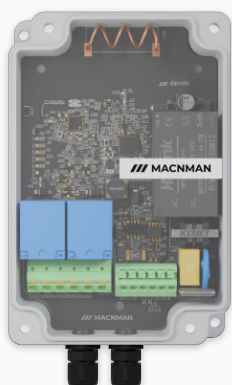
Product Overview

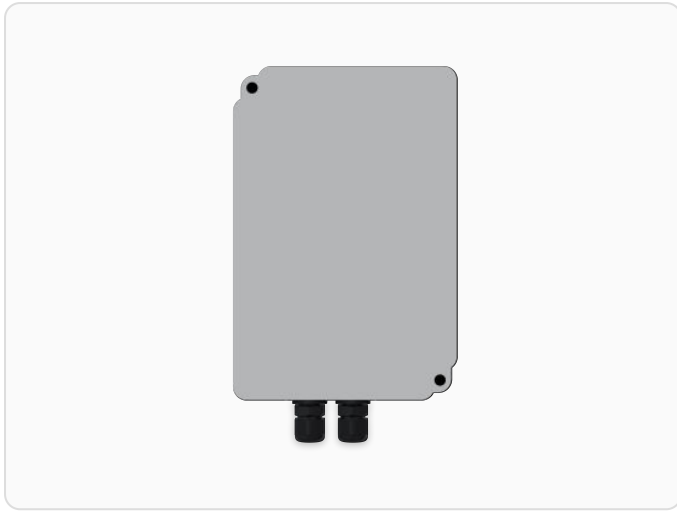
MacSet-WX1 is a robust, power-operated Wi-Fi IoT controller designed for reliable remote control and automation of industrial, commercial, and building infrastructure. Equipped with 2 high-current digital outputs, it enables secure switching of pumps, motors, valves, alarms, solenoids, lighting systems, and other electrical loads over standard Wi-Fi networks.

Designed for demanding environments, MacSet-WX1 features a stable 12–24 V DC power architecture, noise-immune circuitry, and an IP-rated industrial enclosure suitable for continuous operation. By leveraging Wi-Fi connectivity, the controller can communicate directly with on-premise servers, local dashboards, or cloud platforms without the need for gateways.

The device supports MQTT, HTTP/HTTPS communication, along with OTA configuration and firmware updates. Local automation logic allows actions to be executed based on timers, inputs, or rules even when internet connectivity is unavailable.

Ideal for smart buildings, industrial automation, energy management, and remote equipment control, MacSet-WX1 delivers a scalable, gateway-free, and future-ready Wi-Fi control solution.





Key Features

Dual High-Current Digital Outputs

Enables remote switching of pumps, motors, solenoids, alarms, and lighting loads.

Industrial Wi-Fi IoT Controller

Direct IP-based communication over Wi-Fi for local and cloud automation.

Local Automation Logic

Execute actions based on triggers, timers, or inputs—without cloud dependency.

Secure Remote Switching

Encrypted Wi-Fi communication ensures secure and reliable equipment control.

IP-Rated Industrial Enclosure

Rugged design suitable for outdoor, dusty, and moisture-prone environments.

Fail-Safe Output Handling

Supports power-restore behavior, default states, and safe automation routines.

Industrial 12–24 V Power Operation

Stable, noise-immune power input designed for factories and field installations.

Local & Cloud Control Modes

Operate devices via local network control or cloud-based dashboards.

Standard IP Protocols

Supports MQTT, HTTP/HTTPS, and TCP/IP for easy server and cloud integration.

OTA Configuration & Firmware Updates

Remote device configuration and software upgrades via Wi-Fi.

Event-Based Actuation

Trigger outputs based on network commands, schedules, or device rules.

Flexible Platform Integration

Compatible with on-premise servers, private clouds, and local dashboards.

Suited Applications

Smart Agriculture & Irrigation

Remotely control irrigation pumps, valves, solenoids, and fertigation systems using LoRaWAN commands.

Industrial Automation

Switch motors, relays, contactors, alarms, and machinery as part of factory automation workflows.

Solar & Energy Systems

Remotely activate inverter controls, battery bank relays, or system bypass/switching mechanisms.

Cold Storage & Food Industry

Switch compressors, fans, alarms, and standby systems using secure long-range LoRaWAN control.

Smart City Deployments

Operate sirens, public lighting, fountains, signage, and remote utility controllers wirelessly.

Water & Wastewater Management

Operate pumps, blowers, aerators, chlorination units, and valve actuators at remote water sites.

Smart Buildings & Facilities

Control lighting circuits, exhaust fans, access gates, HVAC auxiliary systems, and building utilities.

Remote Infrastructure Control

Manage field devices such as telecom tower equipment, remote panels, and unmanned site actuators.

Pipeline & Industrial Utility Networks

Control valves, pumps, pressure release actuators, and emergency shutoff systems through LoRaWAN.

Environmental Monitoring Systems

Trigger fans, blowers, alarms, or auxiliary devices based on sensor-driven automation rules.

Technical Specifications

General Specifications

Part Order Number	MacSet LX1
Product Brand	Macnman Technologies Pvt Ltd

Interfaces Available

High Current Relay Outputs	x2 @16 Amps
RS485 Interface	1 (Switchable)
4-20 mA Analog Channels	x 2 (Switchable)
0-10 V Analog Channels	x 2 (Switchable)
Digital Inputs	x 2 (Switchable)
ADC Resolution	12-bit

Note

The device use same channels for the RS485 & Analog inputs, either can be used at a time.

Network & Communication

Connectivity	Wi-Fi 2.4 GHz Wi-Fi 5 GHz
Network Protocols	MQTT, HTTP, HTTPS
Operating Mode	Local Network / Cloud Connected

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of MacSet-LX1 is being modified

Device Configuration

Configuration	Via Macman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Remote Configuration	JSON Format Data - from server

Supported Platforms

	On-Premise Servers Private Cloud Platforms MQTT Brokers HTTP/HTTPS APIs Local Dashboards & SCADA Systems
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Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C

Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass field Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 μ A in sleep mode ~50 mA during transmission
Input Impedance	>10 M Ω for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS / HTTP
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

Data Encryption	AES-256 end-to-end encryption
Authentication	Secure Wi-Fi authentication and credential-based access
Firmware Protection	Secure boot and signed OTA updates

MacSet-WX1 Variants

VARIANT NAME	POWER SOURCE	RELAY OUTPUTS	SOLAR SUPPORT	EXTERNAL FLASH	IDEAL FOR
MacSet WX1	12–24 V (Power)	2			Controlling Industrial or commercial sites with stable power supply

What's in the Box?

Make sure box contains following items before installations



MacSet



Mounting Clip*



User manual & Warranty Card



IP Extender*



Mounting Screws

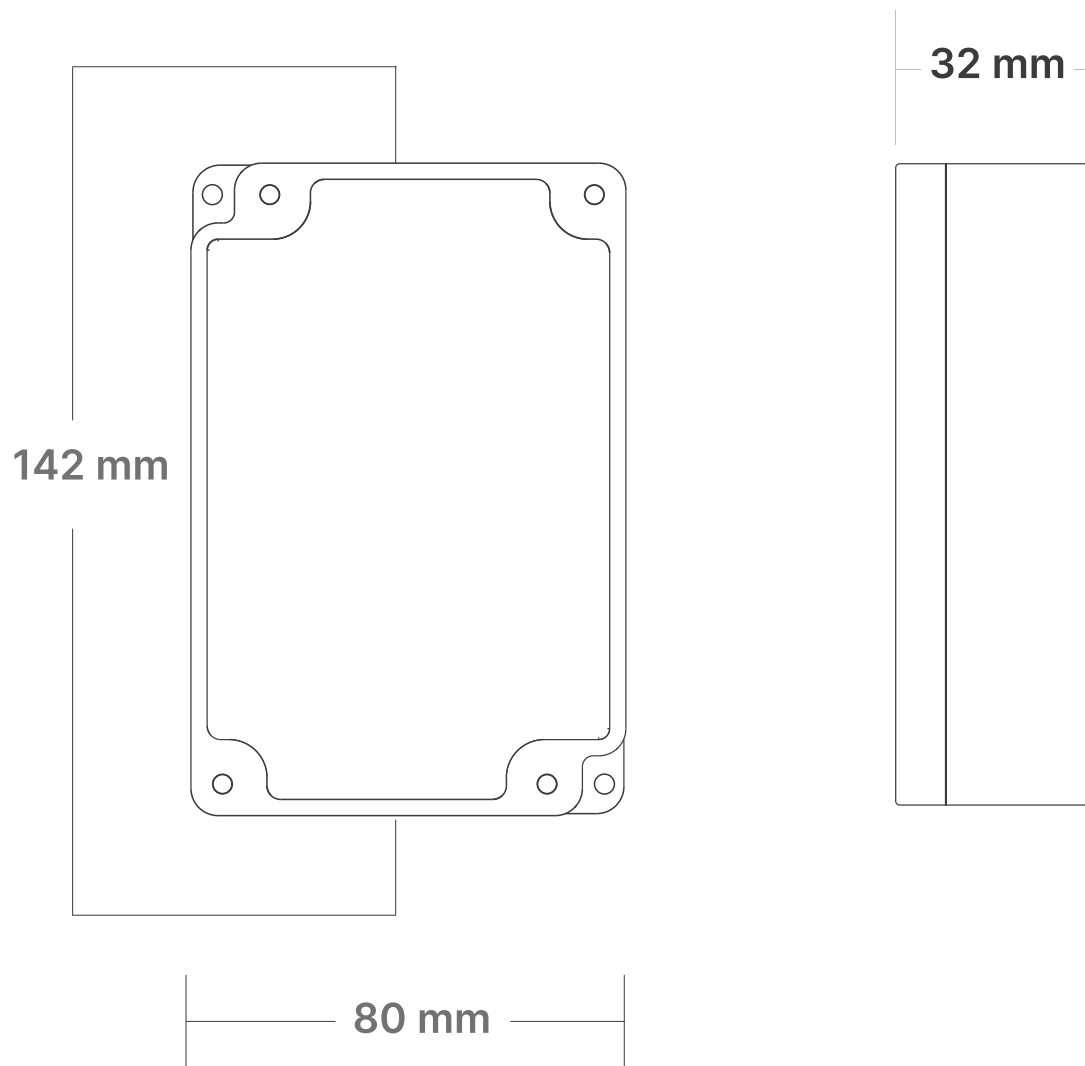


Power Adapters*

Note

Accessories must be purchased separately & may not be included in the box.

MacSet-WX1 Dimensions



ⓘ Note

Accessories must be purchased separately & may not be included in the box.

Frequently Asked Questions

What is the MacSet-WX1 used for?

MacSet-WX1 is a power-operated Wi-Fi IoT controller with 2 high-current digital outputs for remote switching, automation and edge control of equipment. Typical applications include Smart Agriculture & Irrigation, Water & Wastewater Management, Industrial Automation, Smart Buildings & Facilities, Solar & Energy Systems and Remote Infrastructure Control.

What inputs and outputs does the MacSet-WX1 have?

The MacSet-WX1 provides the following interfaces:

- High Current Relay Outputs: x2 @16 Amps
- RS485 Interface: 1 (Switchable)
- 4-20 mA Analog Channels: x 2 (Switchable)
- 0-10 V Analog Channels: x 2 (Switchable)
- Digital Inputs: x 2 (Switchable)
- ADC Resolution: 12-bit

How is the MacSet-WX1 powered?

The MacSet-WX1 is powered from 9-36 V DC (External Supply).

What is the IP rating and operating temperature of the MacSet-WX1?

The MacSet-WX1 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacSet-WX1 configured?

The MacSet-WX1 is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Related Products and Resources

- Related datasheets: [MacSet LX1 industrial LoRaWAN relay controller](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)